

Networking Basics for Cyber Security

Networking is important in cyber security because attacks, defenses, monitoring, and data theft all happen through networks.

Network (What is a Network?)

A **network** is a group of devices connected together to share data.

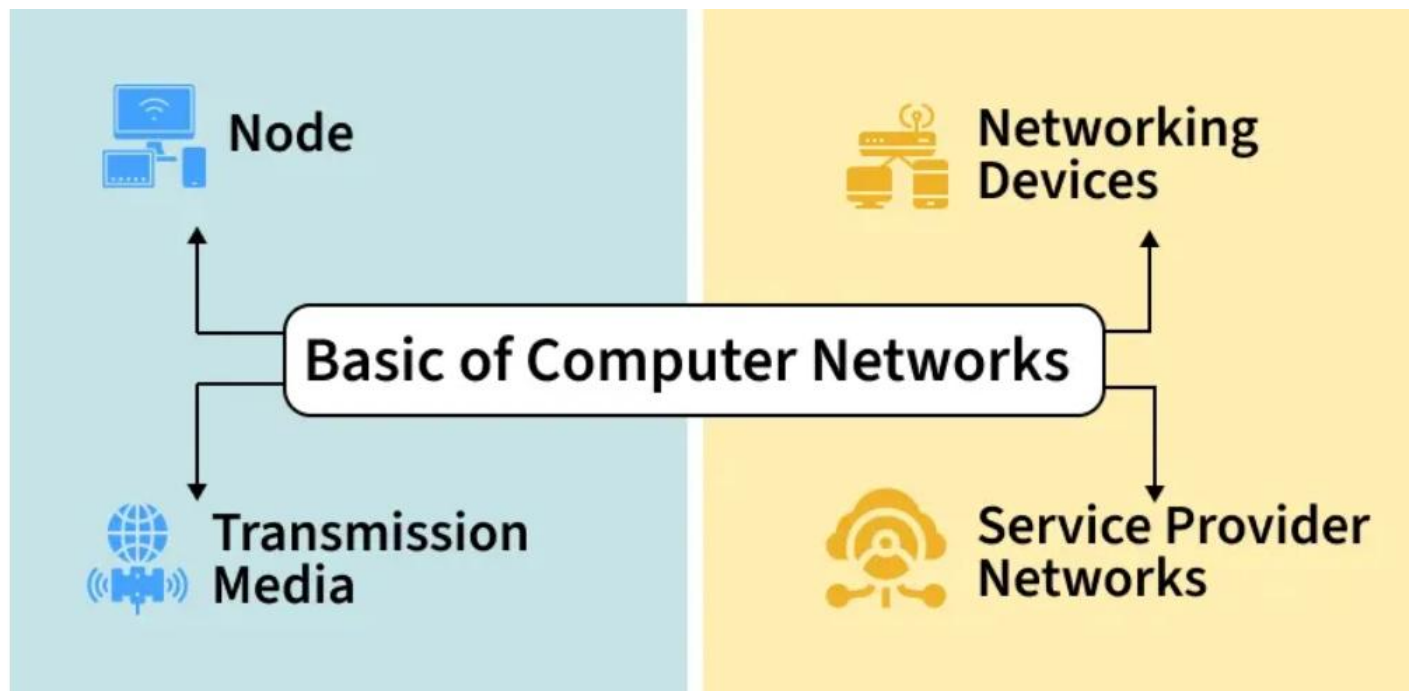
Example

- Mobile phone + Laptop + Wi-Fi router = Network

Basic Terminologies

Basic terminologies of computer networking are:

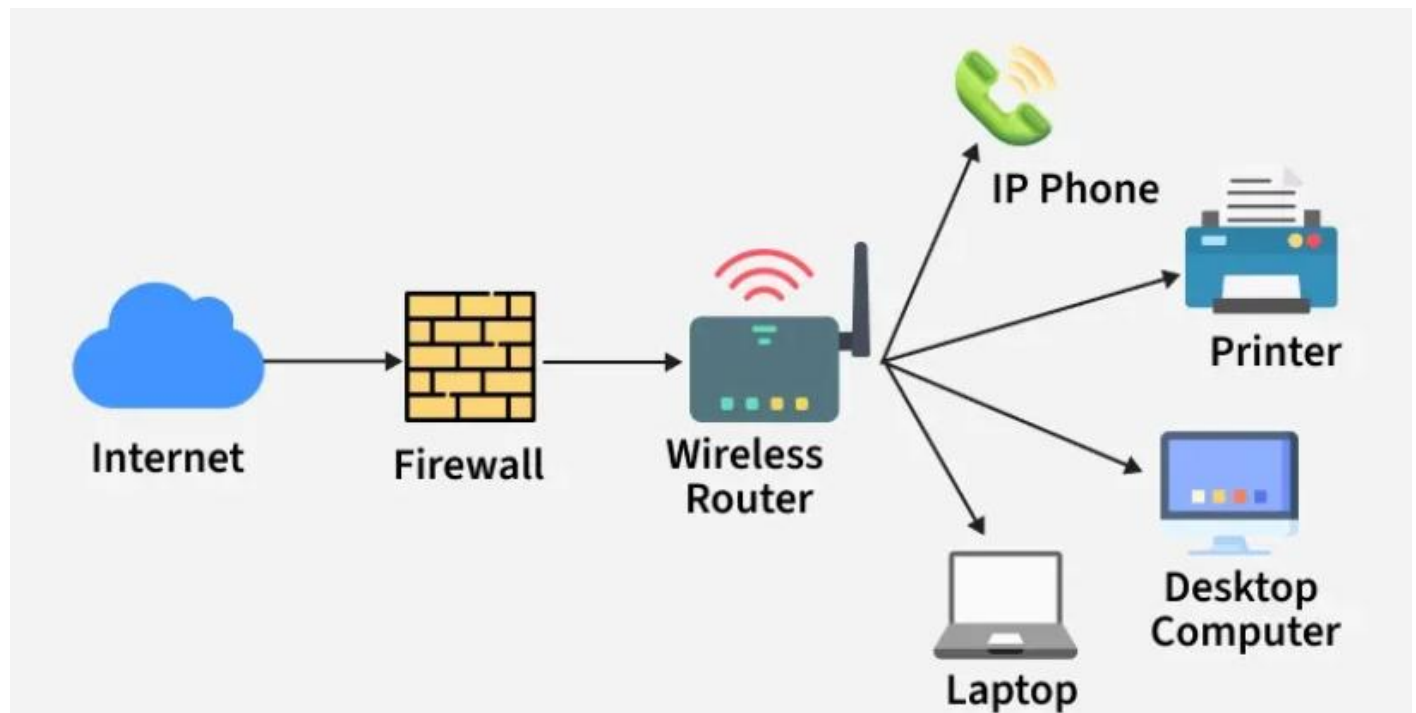
- **Network:** A group of connected computers and devices that can communicate and share data.
- **Node:** Any device that can send, receive, or forward data in a network. This includes laptops, mobiles, printers, earbuds, servers, etc.
- **Networking Devices:** Devices that manage and support networking functions. This includes routers, switches, hubs, and access points.
- **Transmission Media:** The physical or wireless medium through which data travels between devices.
- **Wired media:** Ethernet cables, optical fibre.
- **Wireless media:** Wi-Fi, Bluetooth, infrared
- **Service Provider Networks:** Networks offered by external providers that allow users or organisations to lease network access and capabilities. This includes internet providers, mobile carriers, etc.



Working Of a Computer Network

The basic building blocks of a Computer network are Nodes and Links.

- **Network Node:** Any device in a network, such as a modem, router, or computer, that sends or receives data.
- **Link:** The medium that connects nodes, like cables, wires, or wireless signals.
- **Working:** Networks use rules called protocols to send and receive data between devices. Each device is identified by a unique IP Address.
- **Firewall:** A security tool (hardware or software) that monitors traffic. Based on set rules, it either allows, blocks, or drops data.



Working Structure

Protocols and Networking

It is essential for Cyber Security Professionals to have a solid understanding of how computers communicate. There is much more happening behind the scenes of computer networks than what can be observed when using applications.

The OSI Model

The OSI ("Open Systems Interconnection") model represents an easy and intuitive way to standardize the different parts required to communicate across networks.

Layer	What it does
7 - Application	Where humans process data and information
6 - Presentation	Ensures data is in a usable format
5 - Session	Capable of maintaining connections
4 - Transport	Data is forwarded to a service capable of handling requests
3 - Network Layer	Responsible for which path packets should travel on a network
2 - Data Link	Responsible for which physical devices packets should go to
1 - Physical	The physical infrastructure to transport data

IP - The Internet Protocol

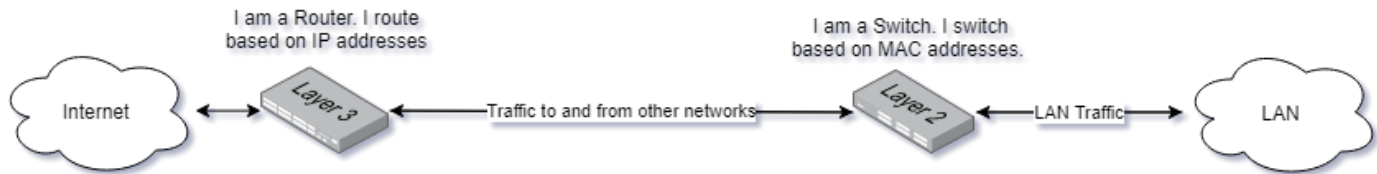
Internet Protocol (IP) is a set of rules used to **identify devices** and **deliver data** from one device to another over a network or the internet.

IP networks can be broken into different sections, often called subnets. This is accomplished by adding an extra piece of information, together with the IP address, called a *netmask*. The netmask dictates how large a network is and which packet is routed within the network and which should be routed outside of the network.

Netmasks can be represented via decimal numbers or with a slash notation. When using slash notation, the slash follows the systems IP address. Here are some examples:

IP Address	Slash Notation	Netmask
10.0.0.1	/8 - Example: 10.0.0.1/8	255.0.0.0
172.16.1.1	/12 - Example: 172.16.1.1/12	255.240.0.0
192.168.0.1	/16 - Example: 192.168.0.1/16	255.255.0.0
192.168.0.1	/24 - Example: 192.168.0.1/24	255.255.255.0

In IP networks the traffic is routed by a router. A router is a networking device which understands the IP format and can forward packets between networks. This is different than a switch as the switch forwards data within a network, while the router forwards between networks.



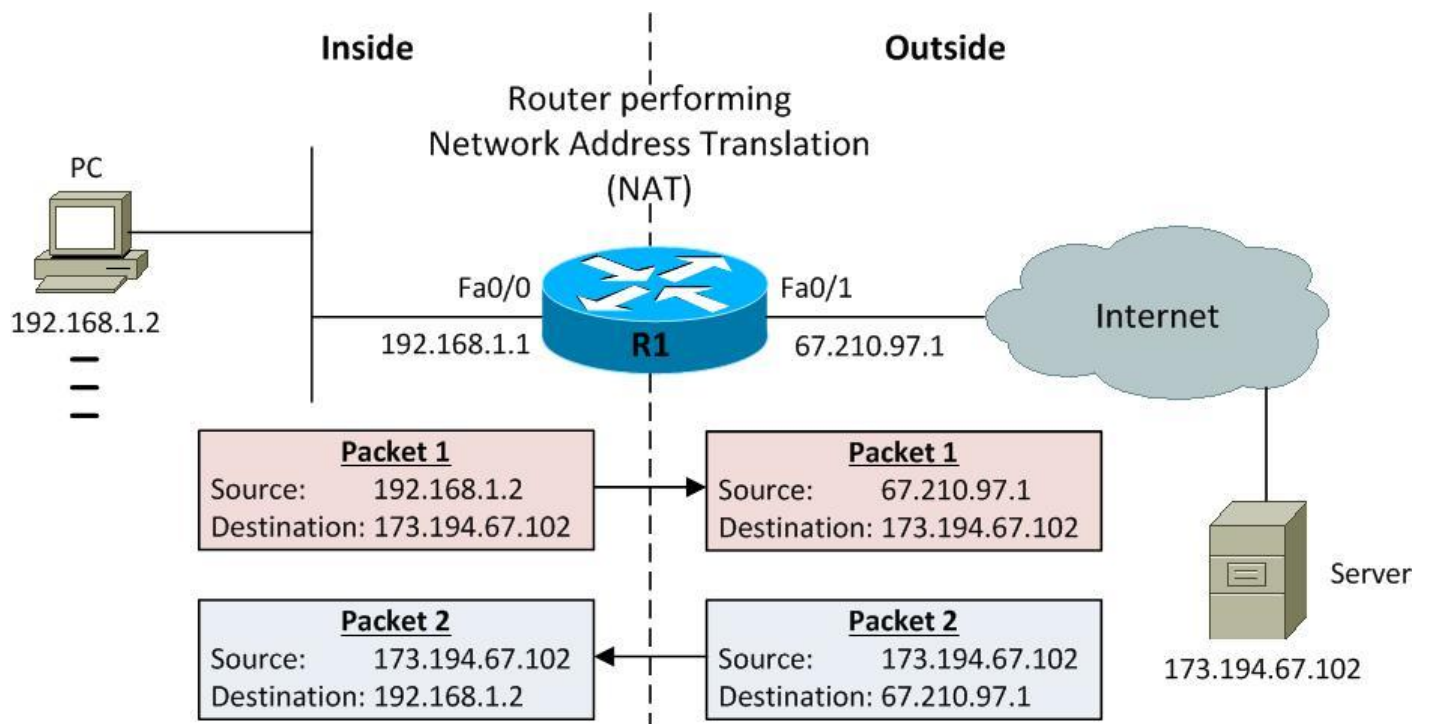
You can check your IP address on Windows by running the command **ipconfig** within a Command Line Window. On Linux this is done with the **ip addr show** or **ifconfig** command.

NAT ("Network Address Translation")

NAT (Network Address Translation) is a technique used by routers to **convert private IP addresses into a public IP address** so devices can access the internet.

NAT allows a system accepting connections on a public IP address to map those requests to an internal RFC 1918 IP address or vice versa. Systems which do NAT'ing are typically firewalls and routers.

A typical implementation of NAT is where the external IP address is used as a front for multiple internal IP addresses, and the destination port number is used to decide which server the data should be sent to. This allows internal IP addresses to receive traffic from external systems.



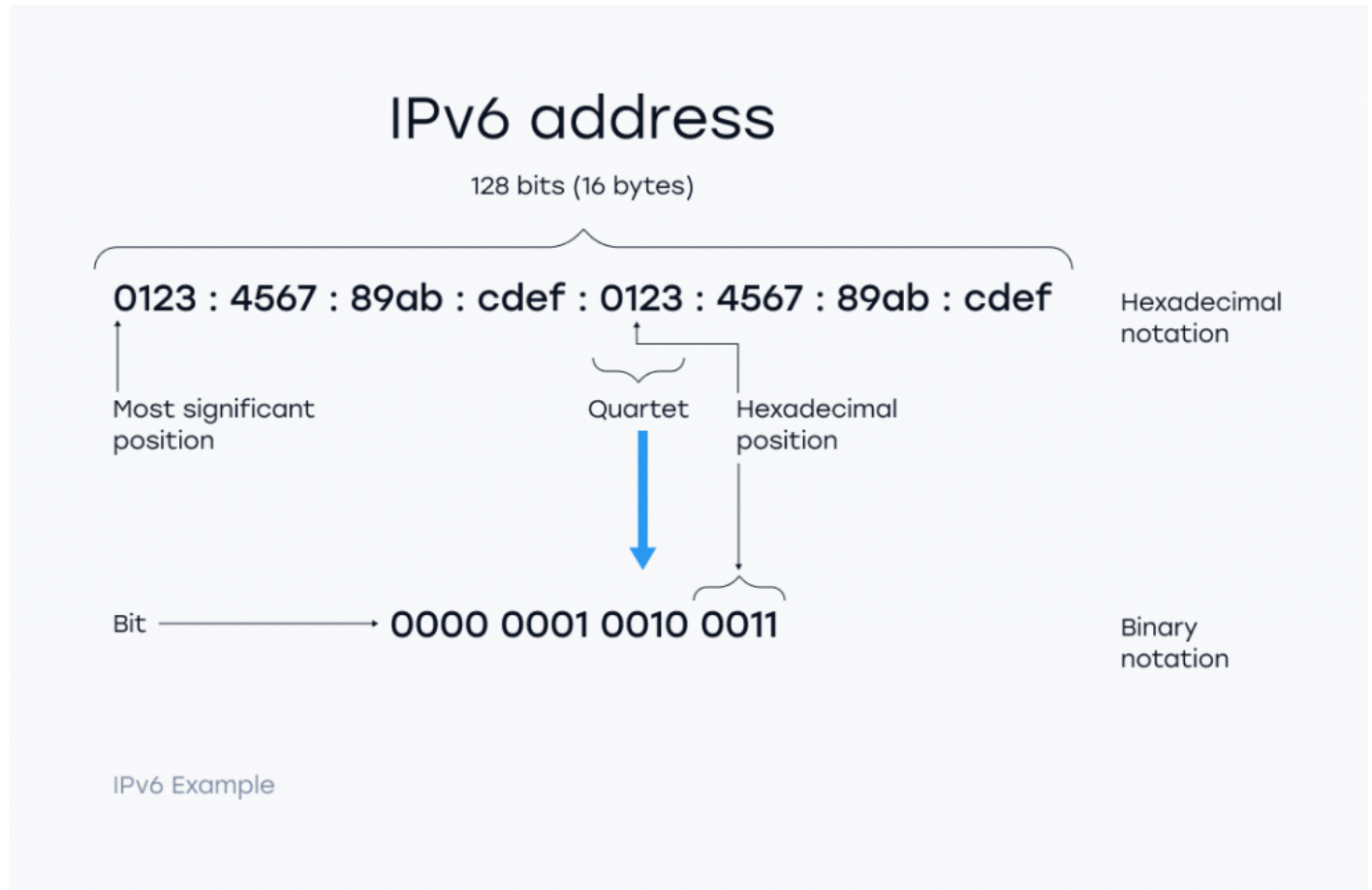
IPv6 - IP Version 6

IP version 6 is the latest standard for IP and was made to support more IP addresses. Instead of using 32 bits of addressing for IP addresses, 128 bits is used. This allows for enough IP addresses for the foreseeable future while IPv4 has already run out.

IPv6 addresses use 8 groups of 4 hexadecimal numbers. An IPv6 address looks like this: 2a00:1450:400f:80a::200e. Notice it does not have the 8 groups of 4 hexadecimal numbers. This is because IPv6 addresses can be shortened via simple rules:

- Leading 0's can be shortened
- Double colon (::) can be used to represent a continuous string of 0's.

The expanded IPv6 address is: 2a00:1450:400f:080a:0000:0000:0000:200e



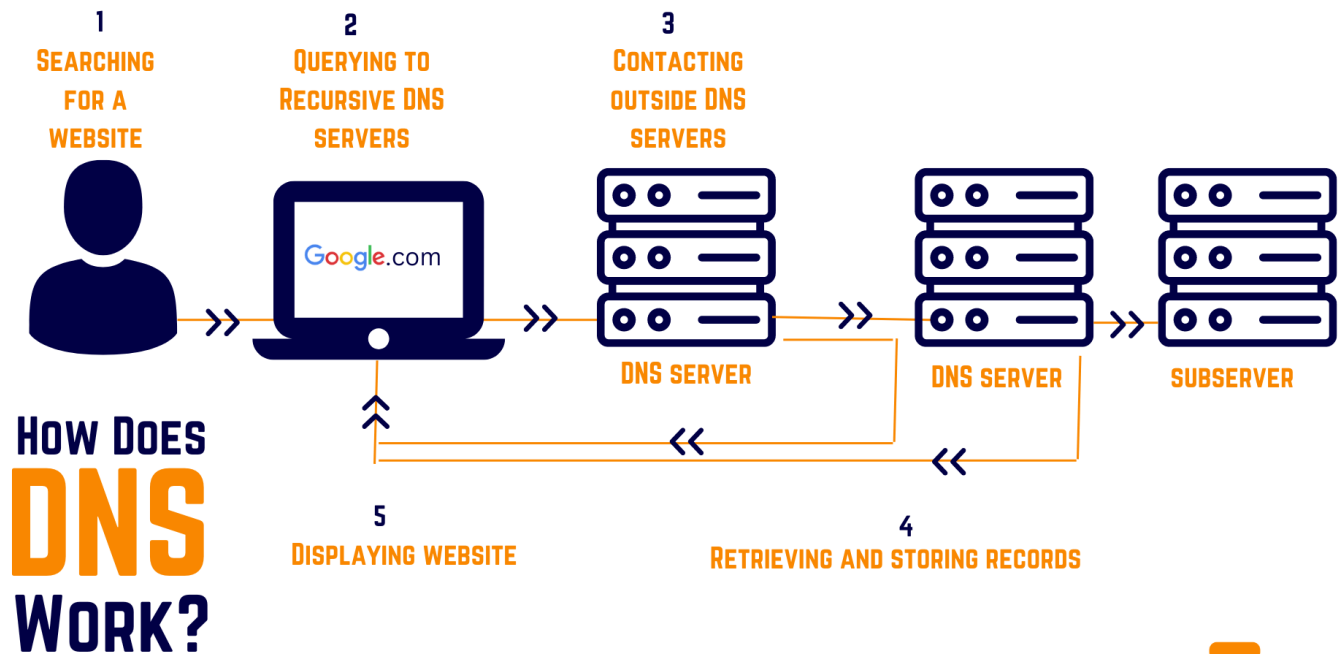
DNS (Domain Name System)

DNS is used to map applications, via names, to IP addresses. For example if you want to use your browser to visit <http://google.com>, the browser must first make a request to a DNS server to resolve the IP address behind google.com

Systems are typically configured with a primary and secondary domain name server. These settings can be configured manually or be provided by a DHCP server. This allows our computer systems to reach a DNS server for it to resolve for us.

The DNS server is then responsible for resolving the request. It will then proceed to check its own cache to see if it already knows the answer. Each DNS answer can be cached, that is stored temporarily to speed up future requests, for a certain TTL ("Time To Live"). The TTL is typically set to a couple of minutes, for example 10 minutes.

If a DNS server does not have an answer in its cache it will then proceed to check who is responsible for giving the answer. This is done via a recursive process which involves asking a hierarchical system of name servers which inevitably will make the DNS request to end up at the Authoritative Name Server.

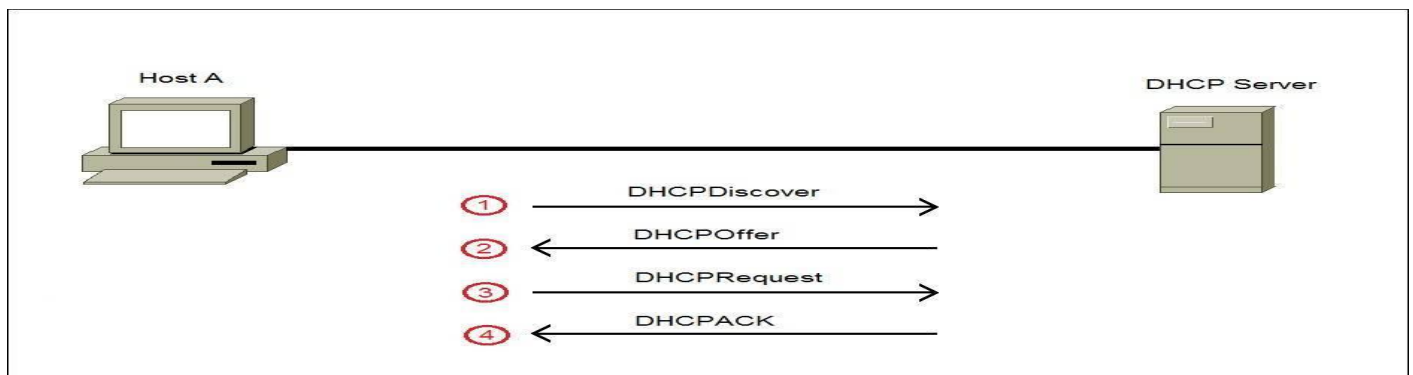


DHCP (Dynamic Host Configuration Protocol)

As the name implies, the DHCP protocol allows any system on a network to reach out to a server and receive a configuration. Such configuration typically implies receiving IP address and network range, default gateway and DNS servers.

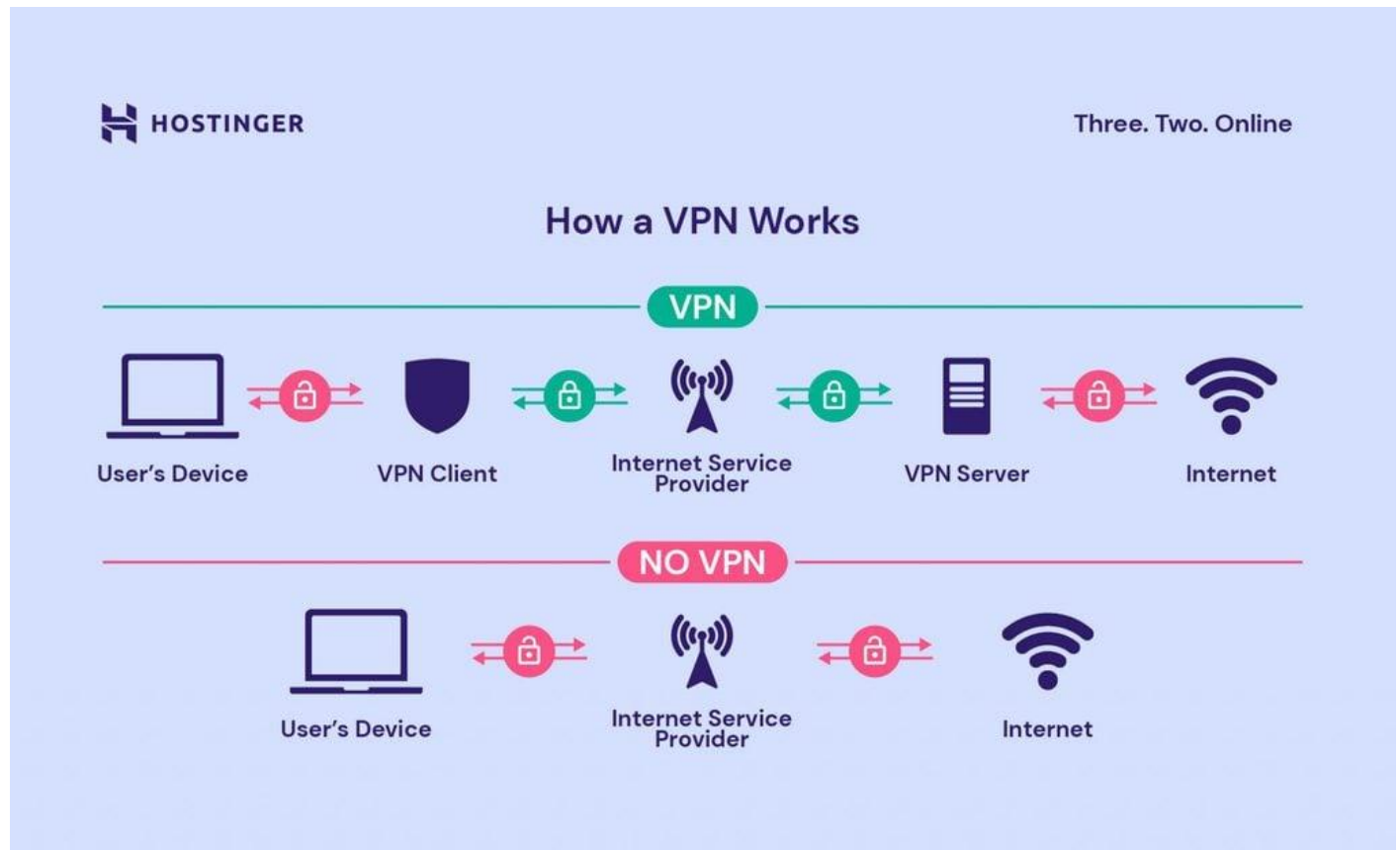
DHCP allows for easy management of clients joining and leaving a network.

If you are curious if you are using DHCP right now you can type `ipconfig /all` on a Windows system and look for "DHCP Enabled: Yes" in the output. Your computer might have multiple network interfaces.



VPN (Virtual Private Network)

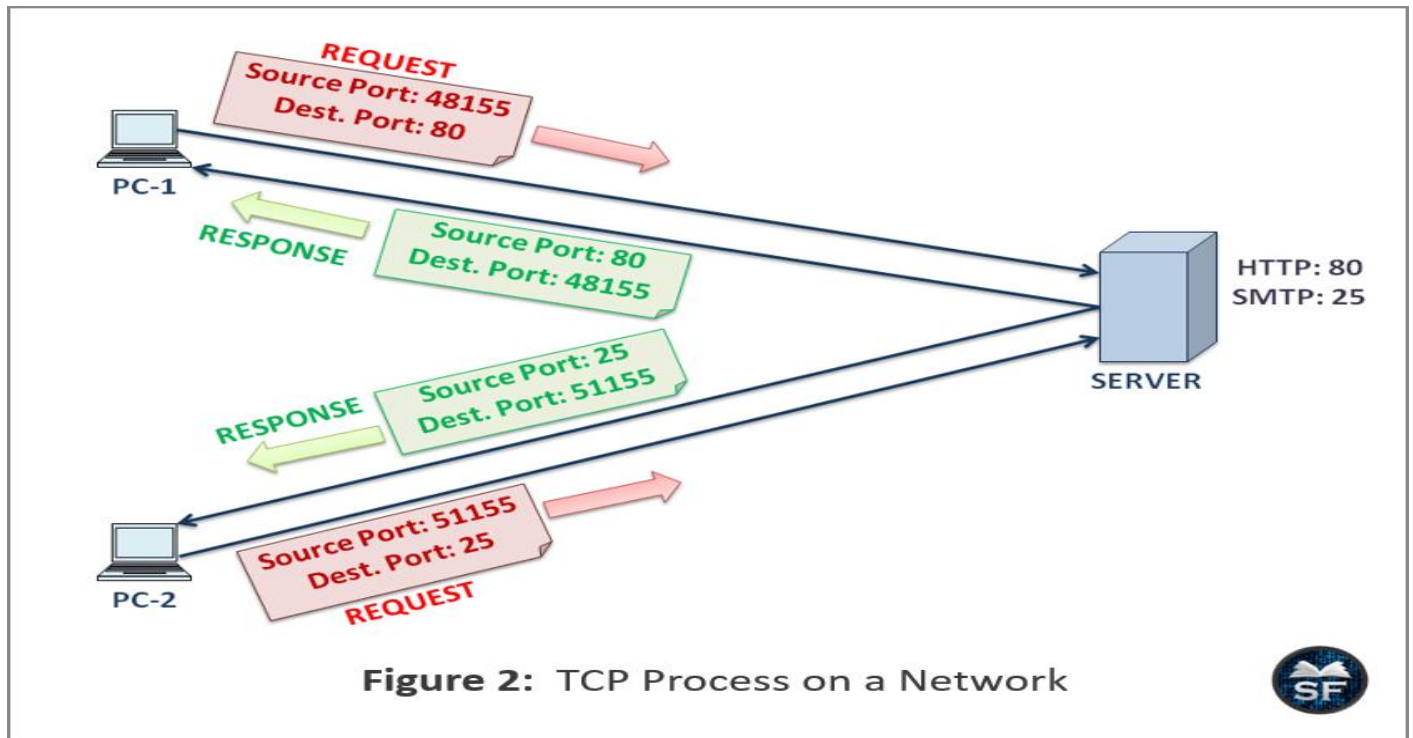
A VPN is a system which enables two system to establish encrypted forms for communication, enabling network traffic to be encrypted in transit. Many VPN's is a client to server architecture, allowing the client to access multiple services behind the VPN. VPN's hosted by your work place is also likely to provide access to resources otherwise only accessible from the inside.



TCP (Transmission Control Protocol)

TCP (Transmission Control Protocol) is a protocol that allows devices to communicate reliably over a network. It ensures that data reaches the destination correctly and in the right order, even if parts of the network are slow or unreliable.

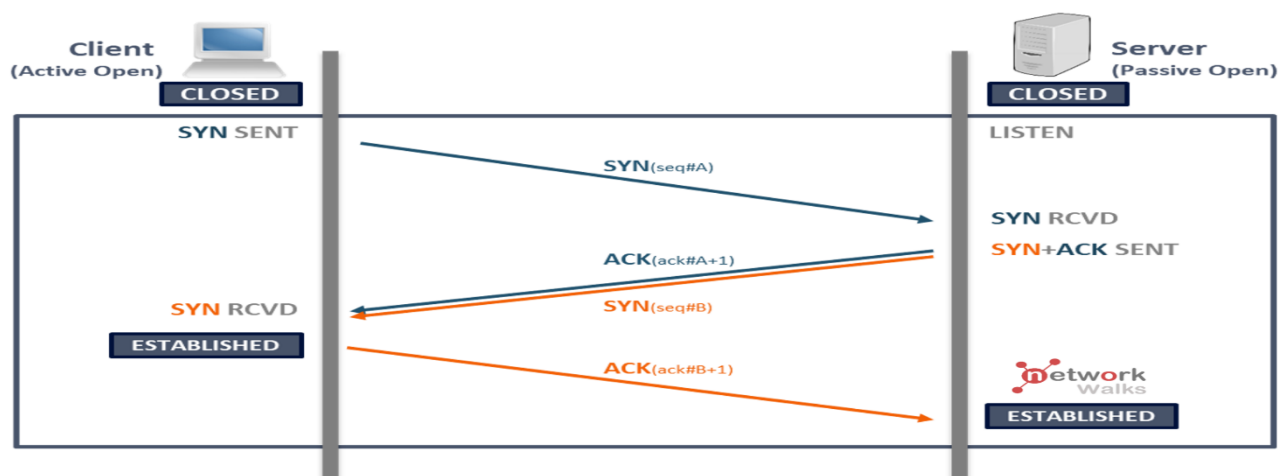
Just like IP uses IP addresses for addressing, TCP and UDP uses ports. A port, as indicated with a number between 0 and 65535, dictates which network service should process the request.



TCP 3-Way-Handshake

TCP uses a three-way handshake to allow two systems to engage in communications. The handshake uses 32 bits of PRNG ("Pseudo Random Number Generator") numbers to establish the handshake. The handshake enforces that both parties intend to communicate.

TCP 3-way Handshake Process



Explanation on how TCP engages in communications:

1. Client initiates the communication by sending a packet with the control bit SYN set in the header, a PRNG number in the Sequence number field and a target destination port. The Network Layer (Layer 3) allows the packet to be sent to a remote system. This packet is referred to as a SYN packet.
2. Server receives the packet, reads the Sequence Number from the Client and crafts a response. The response sets the Acknowledgement field with the Sequencer number of the client with the number 1 added to it. Furthermore the response contains the controls bits SYN and ACK set and the Sequence number is set to the Servers PRNG number. This packet is referred to as a SYN/ACK packet.
3. The Client receives the SYN-ACK packet and to complete the handshake returns a packet with the ACK control bit set. The final packet also confirms the PRNG number from the Server by returning it in the Acknowledgement field of the header.

This process is commonly referred to as the SYN, SYN/ACK, ACK 3-way handshake.

UDP (User Datagram Protocol)

UDP (User Datagram Protocol) is a fast but unreliable transport-layer protocol used for communication where speed is more important than accuracy.

UDP is used for traffic which does not need the resilience and security of TCP, typically applications such as VOIP, but in the modern world more applications are using UDP to support fast packet transfer with resiliency and security built into the higher levels of the OSI model; QUIC is an example of this.

USER DATAGRAM PROTOCOL (UDP)



What is Wireshark?

Wireshark is a network packet analyzer. A network packet analyzer presents captured packet data in as much detail as possible.

You could think of a network packet analyzer as a measuring device for examining what's happening inside a network cable, just like an electrician uses a voltmeter for examining what's happening inside an electric cable (but at a higher level, of course).

In the past, such tools were either very expensive, proprietary, or both. However, with the advent of Wireshark, that has changed. Wireshark is available for free, is open source, and is one of the best packet analyzers available today.